Programming Languages

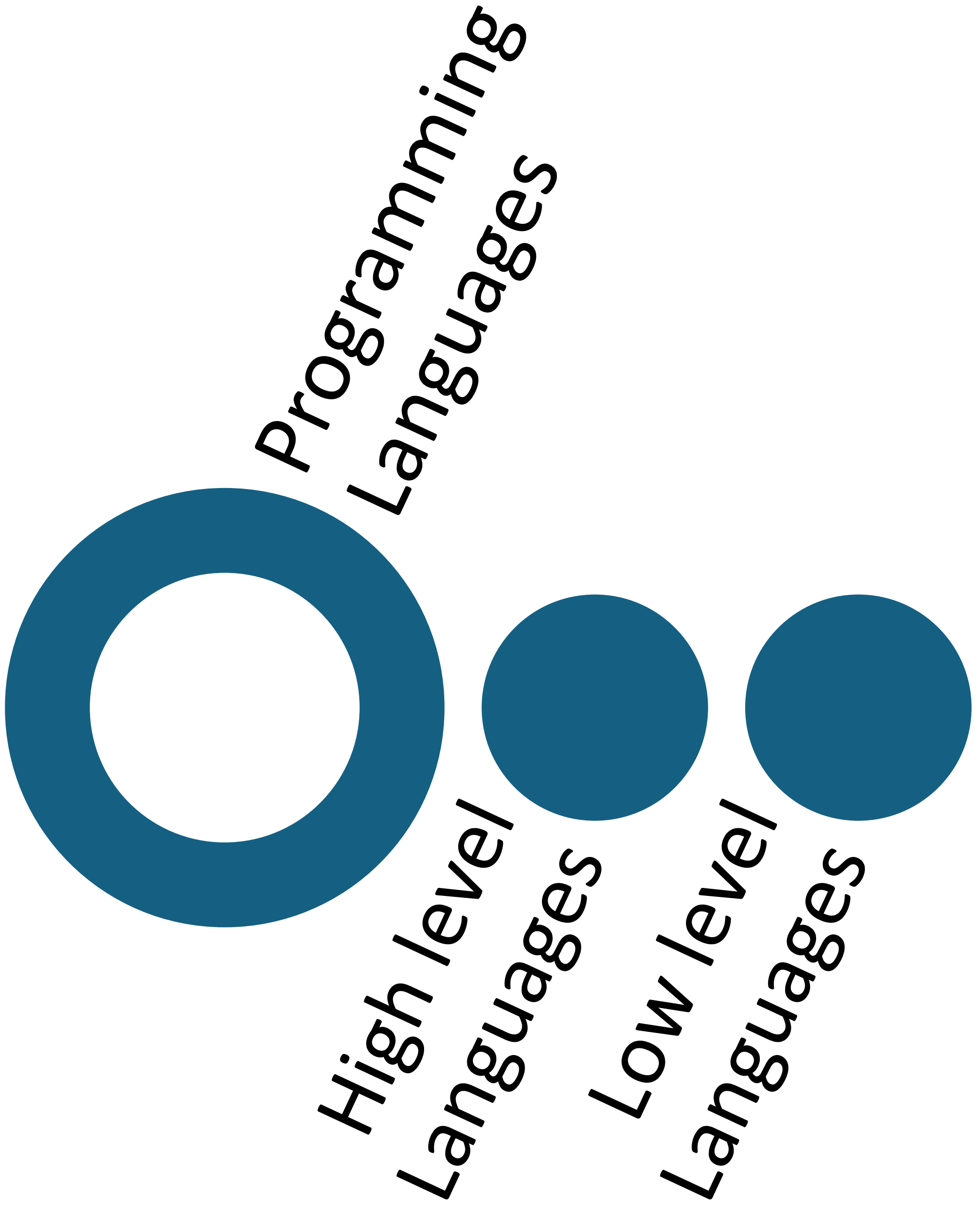
Computer Program

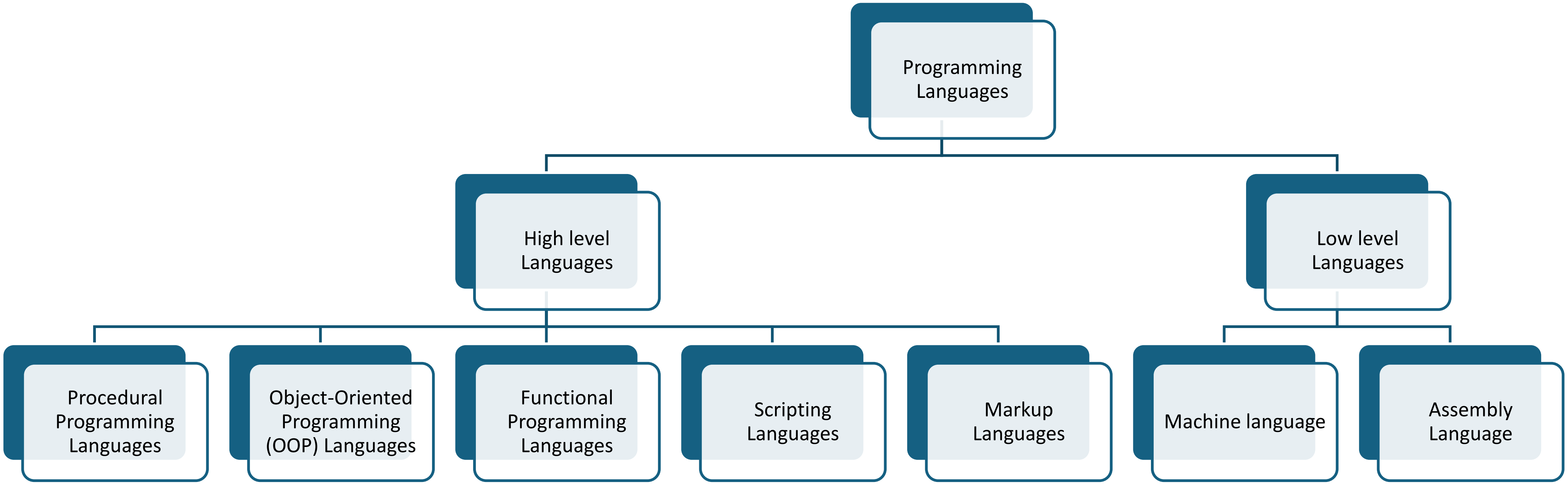
- a program is a set of instructions written in a specific programming language that a computer can understand and execute to perform a particular task or achieve a desired outcome.

- What is an instruction?
- Set of programs??

Programming languages

- A programming language is a formal language designed to communicate instructions to a machine, particularly a computer. It consists of a set of rules that dictate how to write programs, which are sequences of instructions that a computer can execute to perform specific tasks.


Programming Languages
High level Languages
Low level Languages


Programming Languages
High level Languages
Low level Languages
Procedural Programming Languages
Object-Oriented Programming (OOP) Languages
Functional Programming Languages
Scripting Languages
Markup Languages
Machine language
Assembly Language

Low Level Language

Low-Level Languages: These languages are closer to the computer's hardware and require a deep understanding of the machine's architecture.

- **Machine Language:**

This is the lowest-level language, consisting of binary code (0s and 1s) directly understood by the computer's central processing unit (CPU). It is difficult for humans to read and write.

- **Assembly Language:**

A step above machine language, assembly language uses mnemonics (short, symbolic codes) to represent machine instructions. It still requires an assembler to translate it into machine code for execution.

High Level Language

- These languages are more abstract and human-readable, using syntax similar to natural languages. They require a compiler or interpreter to translate them into machine code.


01
Procedural Programming Languages
02
Functional Programming Languages
03
Object-oriented Programming Languages
04
Scripting Languages
05
Logical Programming Languages
06
Imperative Programming

Types of high-level language

- **Procedural Programming Languages:**

These languages focus on a sequence of instructions (procedures or functions) to achieve a task. Examples include C, COBOL, and FORTRAN.

- **Object-Oriented Programming (OOP) Languages:**

These languages organize programs around objects, which are instances of classes that encapsulate data and methods. Examples include Java, Python, C++, and C#.

- **Functional Programming Languages:**

These languages treat computation as the evaluation of mathematical functions and avoid changing state and mutable data. Examples include Haskell and Scala.

Types of high-level language

- **Scripting Languages:**

These are often interpreted languages used to automate tasks, control software applications, or create dynamic web content. Examples include JavaScript, Python, and Ruby.

- **Markup Languages:**

While not traditional programming languages in the sense of executing instructions, they are used to define the structure and presentation of data. Examples include HTML and XML.

Assignment: Investigate and write one pager report on the following subtopics:

- Advantages of high-level language and low-level language
- Concept of translator (Compiler , Assembler, Interpreter)
- What is:
    - Source code
    - Machine code
- Concept of IDE (Integrated development Environment)
- General purpose vs special purpose programming languages
- Deadline: 16th November 2025
- **Don’t Copy paste| Don’t Plagiarize**
- **Plagiarism will lead to cancellation of assignment, and the student will be awarded with Zero marks.**

Types of Programming Languages

- https://www.bbc.co.uk/bitesize/guides/z6x26yc/revision/1#:~:text=Examples%20of%20high%2Dlevel%20languages,and%20machine%20code